PER-FRAME KINEMATIC SOLVE

Frame inputs

Aligned keypoint targets

Previous robot configuration

Warm-start estimate

Evaluate geometry

Site positions + Jacobians

Residuals r(q)

Constraints c(q)

Build convex QP

$H = J^T W J$

Linearize constraints

Bounds on the step Δq

Solve and update

OSQP → step Δq

Optional line search: α

$q \leftarrow q + \alpha \Delta q$

Carry q(t) forward as the initialization for frame t + 1

Objective residuals

Keypoint tracking · nominal joint posture

Temporal smoothing relative to the preceding frame

Explicit constraints

Joint limits · ground clearance · unit root quaternion

Capsule clearance and self-collision when enabled

Initialization uses extra SQP iterations; the default RTI setting uses one SQP step per subsequent frame.